

Solutions to Axler's *Linear Algebra Done Right*, 4th Edition

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6C Orthogonal Complements and Minimization Problems

Exercise 6C1

Suppose $v_1, \dots, v_m \in V$. Prove that

$$\{v_1, \dots, v_m\}^\perp = (\text{span}(v_1, \dots, v_m))^\perp.$$

Solution. Suppose $v \in \{v_1, \dots, v_m\}^\perp$. Then $\langle v, v_k \rangle = 0$ for each $k = 1, \dots, m$. For any $u \in \text{span}(v_1, \dots, v_m)$, we can write $u = a_1 v_1 + \dots + a_m v_m$ for some scalars $a_1, \dots, a_m$. Then

$$\langle v, u \rangle = \sum_{k=1}^m \bar{a}_k \langle v, v_k \rangle = 0,$$

so $v \in (\text{span}(v_1, \dots, v_m))^\perp$.

Conversely, suppose $v \in (\text{span}(v_1, \dots, v_m))^\perp$. Since $\{v_1, \dots, v_m\} \subseteq \text{span}(v_1, \dots, v_m)$, Theorem 6.48(e) shows that $\text{span}(v_1, \dots, v_m)^\perp \subseteq \{v_1, \dots, v_m\}^\perp$, so $v \in \{v_1, \dots, v_m\}^\perp$. ■

Exercise 6C2

Suppose U is a subspace of V with basis $u_1, \dots, u_m$ and

$$u_1, \dots, u_m, v_1, \dots, v_n$$

is a basis of V . Prove that if the Gram–Schmidt procedure is applied to the basis of V above, producing a list $e_1, \dots, e_m, f_1, \dots, f_n$, then $e_1, \dots, e_m$ is an orthonormal basis of U and $f_1, \dots, f_n$ is an orthonormal basis of $U^\perp$.

Solution. Let $e_1, \dots, e_m, f_1, \dots, f_n$ be the result of applying the Gram–Schmidt procedure to the basis $u_1, \dots, u_m, v_1, \dots, v_n$ of V . By the properties of the Gram–Schmidt procedure, $\text{span}(e_1, \dots, e_k) = \text{span}(u_1, \dots, u_k)$ for each $k \leq m$. In particular, $\text{span}(e_1, \dots, e_m) = \text{span}(u_1, \dots, u_m) = U$, so $e_1, \dots, e_m$ is an orthonormal basis of U .

Since each f_j is constructed to be orthogonal to all previous vectors in the Gram–Schmidt process, it follows that each f_j is orthogonal to every e_k for $k = 1, \dots, m$. Hence, $f_1, \dots, f_n \in U^\perp$. Additionally, $f_1, \dots, f_n$ is an orthonormal list in $U^\perp$ because the Gram–Schmidt procedure produces orthonormal vectors. By Theorem 6.49, $V = U \oplus U^\perp$, so $\dim U + \dim U^\perp = \dim V$. Since $\dim U = m$ and $\dim V = m + n$, we have $\dim U^\perp = n$. Therefore, $f_1, \dots, f_n$ is an orthonormal basis of $U^\perp$. ■

Exercise 6C3

Suppose U is the subspace of $\mathbb{R}^4$ defined by

$$U = \text{span}((1, 2, 3, -4), (-5, 4, 3, 2)).$$

Find an orthonormal basis of U and an orthonormal basis of $U^\perp$.

Solution. Let $v_1 = (1, 2, 3, -4)$ and $v_2 = (-5, 4, 3, 2)$. Applying the Gram–Schmidt procedure to v_1 and

v_2 , we first set $f_1 = v_1$ with $\|f_1\|^2 = 30$. Then

$$f_2 = v_2 - \frac{\langle v_2, f_1 \rangle}{\|f_1\|^2} f_1 = (-5, 4, 3, 2) - \frac{4}{30}(1, 2, 3, -4) = \frac{1}{15}(-77, 56, 39, 38),$$

where

$$\|f_2\|^2 = \frac{1}{225}((-77)^2 + 56^2 + 39^2 + 38^2) = \frac{12030}{225}.$$

Then an orthonormal basis of U is given by

$$e_1 = \frac{1}{\sqrt{30}}(1, 2, 3, -4), \quad e_2 = \frac{1}{\sqrt{12030}}(-77, 56, 39, 38).$$

An orthonormal basis for $U^\perp$ consists of vectors orthogonal to both v_1 and v_2 . To find such vectors, consider the system of equations

$$\begin{cases} x + 2y + 3z - 4w = 0 \\ -5x + 4y + 3z + 2w = 0 \end{cases}$$

The first equation yields $x = -2y - 3z + 4w$, and substituting into the second equation gives

$$-5(-2y - 3z + 4w) + 4y + 3z + 2w = 0 \implies 14y + 18z - 18w = 0 \implies 7y = -9z + 9w.$$

Choosing $z = w = 1$ gives $y = 0$, and then $x = -2(0) - 3(1) + 4(1) = 1$. Choosing $z = 0$ and $w = 7$ gives $y = 9$ and then $x = -2(9) - 3(0) + 4(7) = -18 + 28 = 10$. Hence, we have two vectors $u_1 = (1, 0, 1, 1)$ and $u_2 = (10, 9, 0, 7)$ in $U^\perp$. Applying the Gram–Schmidt procedure to u_1 and u_2 , we first set $g_1 = u_1$ with $\|g_1\|^2 = 3$. Then

$$g_2 = u_2 - \frac{\langle u_2, g_1 \rangle}{\|g_1\|^2} g_1 = (10, 9, 0, 7) - \frac{17}{3}(1, 0, 1, 1) = \frac{1}{3}(13, 27, -17, 4),$$

where

$$\|g_2\|^2 = \frac{1}{9}(13^2 + 27^2 + (-17)^2 + 4^2) = \frac{1203}{9}.$$

Then an orthonormal basis of $U^\perp$ is given by

$$e'_1 = \frac{1}{\sqrt{3}}(1, 0, 1, 1), \quad e'_2 = \frac{1}{\sqrt{1203}}(13, 27, -17, 4). \quad \blacksquare$$

Exercise 6C4

Suppose $e_1, \dots, e_n$ is a list of vectors in V with $\|e_k\| = 1$ for each $k = 1, \dots, n$ and

$$\|v\|^2 = |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_n \rangle|^2$$

for all $v \in V$. Prove that $e_1, \dots, e_n$ is an orthonormal basis of V .

Axler Note. This exercise provides a converse to 6.30(b).

Solution. Setting $v = e_j$, we have

$$1 = \|e_j\|^2 = |\langle e_j, e_1 \rangle|^2 + \cdots + |\langle e_j, e_n \rangle|^2 = |\langle e_j, e_j \rangle|^2 + \sum_{k \neq j} |\langle e_j, e_k \rangle|^2.$$

Since $\|e_j\| = 1$, we have $|\langle e_j, e_j \rangle|^2 = 1$, so

$$1 = 1 + \sum_{k \neq j} |\langle e_j, e_k \rangle|^2 \implies \sum_{k \neq j} |\langle e_j, e_k \rangle|^2 = 0,$$

which forces $|\langle e_j, e_k \rangle|^2 = 0$ for each $k \neq j$, so $\langle e_j, e_k \rangle = 0$ for each $j \neq k$. Hence, $e_1, \dots, e_n$ is an orthonormal list. To show that $e_1, \dots, e_n$ is a basis of V , let $v \in V$ be arbitrary. Consider

$$u = v - \sum_{k=1}^n \langle v, e_k \rangle e_k.$$

Then for each j :

$$\langle u, e_j \rangle = \langle v, e_j \rangle - \sum_{k=1}^n \langle v, e_k \rangle \langle e_k, e_j \rangle = \langle v, e_j \rangle - \langle v, e_j \rangle = 0$$

by orthonormality. Thus, u is orthogonal to each e_j , so by the given condition,

$$\|u\|^2 = |\langle u, e_1 \rangle|^2 + \cdots + |\langle u, e_n \rangle|^2 = 0,$$

which implies $u = 0$. Therefore, $v \in \text{span}(e_1, \dots, e_n)$. Since $e_1, \dots, e_n$ is an orthonormal list and $\text{span}(e_1, \dots, e_n) = V$, it follows that $e_1, \dots, e_n$ is an orthonormal basis of V . ■

Exercise 6C5

Suppose that V is finite-dimensional and U is a subspace of V . Show that $P_{U^\perp} = I - P_U$, where I is the identity operator on V .

Solution. For every $v \in V$, $v = P_U v + P_{U^\perp} v$ by the definition of orthogonal projection. Then

$$P_{U^\perp} v = v - P_U v = Iv - P_U v = (I - P_U)v.$$

Hence, $P_{U^\perp} = I - P_U$. ■

Exercise 6C6

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Show that

$$T = TP_{(\text{null } T)^\perp} = P_{\text{range } T}T.$$

Solution. We first show that $T = TP_{(\text{null } T)^\perp}$. For any $v \in V$, write $v = u + w$ for some $u \in \text{null } T$ and $w \in (\text{null } T)^\perp$. Then

$$TP_{(\text{null } T)^\perp} v = TP_{(\text{null } T)^\perp}(u + w) = TP_{(\text{null } T)^\perp}u + TP_{(\text{null } T)^\perp}w = T0 + Tw = Tw.$$

Since $u \in \text{null } T$, we have $Tu = 0$, so $Tv = T(u + w) = Tu + Tw = 0 + Tw = Tw$. Hence, $T = TP_{(\text{null } T)^\perp}$.

Next, we show that $T = P_{\text{range } T}T$. For any $v \in V$, we have $Tv \in \text{range } T$, so $P_{\text{range } T}Tv = Tv$. Therefore, $T = P_{\text{range } T}T$. ■

Exercise 6C7

Suppose that X and Y are finite-dimensional subspaces of V . Prove that $P_X P_Y = 0$ if and only if $\langle x, y \rangle = 0$ for all $x \in X$ and all $y \in Y$.

Solution. $\langle x, y \rangle = 0$ for all $x \in X, y \in Y \iff Y \subseteq X^\perp \iff \text{range } P_Y \subseteq \text{null } P_X \iff P_X P_Y = 0$. ■

Exercise 6C8

Suppose U is a finite-dimensional subspace of V and $v \in V$. Define a linear functional $\varphi: U \rightarrow \mathbb{F}$ by

$$\varphi(u) = \langle u, v \rangle$$

for all $u \in U$. By the Riesz representation theorem (6.42) as applied to the inner product space U , there exists a unique vector $w \in U$ such that

$$\varphi(u) = \langle u, w \rangle$$

for all $u \in U$. Show that $w = P_U v$.

Solution. By the Riesz representation theorem, we have $w \in U$. For any $u \in U$, we have

$$\langle u, v - w \rangle = \langle u, v \rangle - \langle u, w \rangle = \varphi(u) - \varphi(u) = 0.$$

Hence, $v - w \in U^\perp$, so $w = P_U v$. ■

Exercise 6C9

Suppose V is finite-dimensional. Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$ and every vector in $\text{null } P$ is orthogonal to every vector in $\text{range } P$. Prove that there exists a subspace U of V such that $P = P_U$.

Solution. Let $U = \text{range } P$. By Exercise 3B27, $P^2 = P$ implies $V = U \oplus \text{null } P$. Since every $v \in \text{null } P$ is orthogonal to every $u \in U$, we have $\text{null } P \subseteq U^\perp$. From the fundamental theorem of linear maps,

$$\dim \text{null } P = \dim V - \dim U = \dim U^\perp.$$

Hence, $\text{null } P = U^\perp$, so $V = U \oplus U^\perp$. Then for any $x \in V$, we can write $x = u + w$ for some $u \in U$ and $w \in U^\perp$. Since $u \in \text{range } P$ and $P^2 = P$, we have $Pu = u$. Then $Px = P(u + w) = Pu + Pw = u + 0 = u = P_U x$, so $P = P_U$. ■

Exercise 6C10

Suppose V is finite-dimensional and $P \in \mathcal{L}(V)$ is such that $P^2 = P$ and

$$\|Pv\| \leq \|v\|$$

for every $v \in V$. Prove that there exists a subspace U of V such that $P = P_U$.

Solution. Let $U = \text{range } P$. From Exercise 3B27, $P^2 = P$ implies $V = U \oplus \text{null } P$. Let $w \in \text{null } P$ and $u \in U$. For every $\alpha \in \mathbb{F}$,

$$\|u\| = \|P(u + \alpha w)\| \leq \|u + \alpha w\|.$$

By Exercise 6A6, $\langle u, w \rangle = 0$. Therefore, $\text{null } P \subseteq U^\perp$. From the fundamental theorem of linear maps,

$$\dim \text{null } P = \dim V - \dim U = \dim U^\perp.$$

Hence, $\text{null } P = U^\perp$, so $V = U \oplus U^\perp$. For any $v \in V$, write $v = u + w$ with $u \in U$ and $w \in U^\perp = \text{null } P$. Then $Pv = Pu + Pw = Pu = u = P_U v$, where $Pu = u$ since $u \in \text{range } P$ and $P^2 = P$. Hence, $P = P_U$. ■

Exercise 6C11

Suppose $T \in \mathcal{L}(V)$ and U is a finite-dimensional subspace of V . Prove that

$$U \text{ is invariant under } T \iff P_U T P_U = T P_U.$$

Solution. Suppose U is invariant under T . For any $v \in V$, we have $P_U v \in U$, so $T P_U v \in U$ by invariance. Since P_U acts as the identity on U , it follows that $P_U T P_U v = T P_U v$.

Conversely, suppose $P_U T P_U = T P_U$. For any $u \in U$, we have $P_U u = u$, so

$$Tu = T P_U u = P_U T P_U u \in \text{range } P_U = U,$$

so U is invariant under T . ■

Exercise 6C12

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and U is a subspace of V . Prove that

$$U \text{ and } U^\perp \text{ are both invariant under } T \iff P_U T = T P_U.$$

Solution. Suppose U and $U^\perp$ are both invariant under T . For any $v \in V$, write $v = u + w$ with $u \in U$ and $w \in U^\perp$. Then

$$P_U T v = P_U (Tu + Tw) = P_U Tu + P_U Tw = Tu + 0 = Tu,$$

since $Tu \in U$ and $Tw \in U^\perp$. On the other hand,

$$T P_U v = T P_U (u + w) = Tu,$$

so $P_U T v = T P_U v$ for all $v \in V$, and hence $P_U T = T P_U$.

Conversely, suppose $P_U T = T P_U$. For any $u \in U$, we have $P_U (Tu) = T(P_U u) = Tu$, so $Tu \in U$, and U is invariant under T . For any $w \in U^\perp$, we have $P_U (Tw) = T(P_U w) = T(0) = 0$, so $Tw \in U^\perp$, and $U^\perp$ is invariant under T . ■

Exercise 6C13

Suppose $\mathbb{F} = \mathbb{R}$ and V is finite-dimensional. For each $v \in V$, let φ_v denote the linear functional on V defined by

$$\varphi_v(u) = \langle u, v \rangle$$

for all $u \in V$.

- (a) Show that $v \mapsto \varphi_v$ is an injective linear map from V to V' .
- (b) Use (a) and a dimension-counting argument to show that $v \mapsto \varphi_v$ is an isomorphism from V onto V' .

Axler Note. The purpose of this exercise is to give an alternative proof of the Riesz representation theorem (6.42 and 6.58) when $\mathbb{F} = \mathbb{R}$. Thus, you should not use the Riesz representation theorem as a tool in your solution.

Solution.

- (a) Define the map $\Phi : V \rightarrow V'$ by $\Phi(v) = \varphi_v$ for all $v \in V$. To show that Φ is linear, let $v_1, v_2 \in V$ and $a, b \in \mathbb{R}$. Then for any $u \in V$,

$$\begin{aligned} \Phi(av_1 + bv_2)(u) &= \varphi_{av_1 + bv_2}(u) \\ &= \langle u, av_1 + bv_2 \rangle \\ &= a\langle u, v_1 \rangle + b\langle u, v_2 \rangle \\ &= a\varphi_{v_1}(u) + b\varphi_{v_2}(u) \\ &= a\Phi(v_1)(u) + b\Phi(v_2)(u). \end{aligned}$$

Since this holds for all $u \in V$, we have $\Phi(av_1 + bv_2) = a\Phi(v_1) + b\Phi(v_2)$, so Φ is linear. Injectivity follows from the fact that if $\Phi(v) = 0$, then $\varphi_v(u) = 0$ for all $u \in V$, which implies $\langle u, v \rangle = 0$ for all $u \in V$. In particular, taking $u = v$ gives $\langle v, v \rangle = 0$, so $v = 0$.

- (b) By part (a), Φ is an injective linear map from V to V' . Since V is finite-dimensional, we have $\dim V = \dim V'$. Hence, Φ is also surjective, and therefore an isomorphism from V onto V' . ■

Exercise 6C14

Suppose that $e_1, \dots, e_n$ is an orthonormal basis of V . Explain why the dual basis (see 3.112) of $e_1, \dots, e_n$ is $e_1, \dots, e_n$ under the identification of V' with V provided by the Riesz representation theorem (6.58).

Solution. Let $\varphi_1, \dots, \varphi_n$ be the dual basis of $e_1, \dots, e_n$. By the Riesz representation theorem, for each j there exists a unique $v_j \in V$ such that $\varphi_j(u) = \langle u, v_j \rangle$ for all $u \in V$. Define $\psi_j \in V'$ by $\psi_j(u) = \langle u, e_j \rangle$. Then for any k ,

$$\psi_j(e_k) = \langle e_k, e_j \rangle = \delta_{jk} = \varphi_j(e_k).$$

Since ψ_j and φ_j agree on a basis of V , they are equal. By uniqueness in the Riesz representation theorem, $v_j = e_j$. Hence, the dual basis $\varphi_1, \dots, \varphi_n$ corresponds to $e_1, \dots, e_n$ under the identification of V' with V provided by the Riesz representation theorem. ■

Exercise 6C15

In $\mathbb{R}^4$, let

$$U = \text{span}((1, 1, 0, 0), (1, 1, 1, 2)).$$

Find $u \in U$ such that $\|u - (1, 2, 3, 4)\|$ is as small as possible.

Solution. Applying the Gram–Schmidt procedure to the basis $v_1 = (1, 1, 0, 0)$ and $v_2 = (1, 1, 1, 2)$ of U , we first set $f_1 = v_1$ with $\|f_1\|^2 = 2$. Then

$$f_2 = v_2 - \frac{\langle v_2, f_1 \rangle}{\|f_1\|^2} f_1 = (1, 1, 1, 2) - \frac{2}{2}(1, 1, 0, 0) = (0, 0, 1, 2),$$

with $\|f_2\|^2 = 5$. Projecting $(1, 2, 3, 4)$ onto U gives

$$P_U(1, 2, 3, 4) = \frac{\langle (1, 2, 3, 4), f_1 \rangle}{\|f_1\|^2} f_1 + \frac{\langle (1, 2, 3, 4), f_2 \rangle}{\|f_2\|^2} f_2 = \frac{3}{2}(1, 1, 0, 0) + \frac{11}{5}(0, 0, 1, 2) = \left(\frac{3}{2}, \frac{3}{2}, \frac{11}{5}, \frac{22}{5}\right). \blacksquare$$

Exercise 6C16

Suppose $C[-1, 1]$ is the vector space of continuous real-valued functions on the interval $[-1, 1]$ with inner product given by

$$\langle f, g \rangle = \int_{-1}^1 fg$$

for all $f, g \in C[-1, 1]$. Let U be the subspace of $C[-1, 1]$ defined by

$$U = \{f \in C[-1, 1] \mid f(0) = 0\}.$$

(a) Show that $U^\perp = \{0\}$.

(b) Show that 6.49 and 6.52 do not hold without the finite-dimensional hypothesis.

Appendix. See Appendix 6C16 for an analysis-focused solution to part (a).

Solution.

(a) Let $g \in U^\perp$, and let $f(x) = x^2g(x)$. Then f is continuous on $[-1, 1]$ and $f(0) = 0$, so $f \in U$. Hence,

$$0 = \langle f, g \rangle = \int_{-1}^1 x^2g(x)^2 dx.$$

Since $x^2g(x)^2 \geq 0$ for all $x \in [-1, 1]$, it follows that $x^2g(x)^2 = 0$ for all $x \in [-1, 1]$. Therefore, $g(x) = 0$ for all $x \in [-1, 1] \setminus \{0\}$, and by continuity, $g(0) = 0$ as well. Hence, $g = 0$, so $U^\perp = \{0\}$.

(b) 6.49 asserts that $V = U \oplus U^\perp$, but $U^\perp = \{0\}$, so $U \oplus U^\perp = U \neq C[-1, 1]$ since the constant function 1 is not in U . 6.52 asserts that $(U^\perp)^\perp = U$, but $(U^\perp)^\perp = \{0\}^\perp = C[-1, 1] \neq U$. $\blacksquare$

Exercise 6C17

Find $p \in \mathcal{P}_3(\mathbb{R})$ such that $p(0) = 0$, $p'(0) = 0$, and $\int_0^1 |2 + 3x - p(x)|^2 dx$ is as small as possible.

Solution. The conditions $p(0) = 0$ and $p'(0) = 0$ imply that $p(x) = ax^2 + bx^3$ for some scalars $a, b \in \mathbb{R}$. Let $f(x) = 2 + 3x$ and $g(x) = ax^2 + bx^3$. To minimize the integral, we project f onto $\text{span}(x^2, x^3)$ in $C[0, 1]$ to get the resulting system of equations:

$$\begin{aligned} 0 = \langle f - g, x^2 \rangle &= \int_0^1 (2 + 3x - ax^2 - bx^3)x^2 \, dx = \frac{2}{3} + \frac{3}{4} - \frac{a}{5} - \frac{b}{6}, \\ 0 = \langle f - g, x^3 \rangle &= \int_0^1 (2 + 3x - ax^2 - bx^3)x^3 \, dx = \frac{1}{2} + \frac{3}{5} - \frac{a}{6} - \frac{b}{7}. \end{aligned}$$

Solving this system yields $a = 24$ and $b = -\frac{203}{10}$, so the polynomial $p(x) = 24x^2 - \frac{203}{10}x^3$ minimizes the integral. ■

Exercise 6C18

Find $p \in \mathcal{P}_5(\mathbb{R})$ that makes $\int_{-\pi}^{\pi} |\sin x - p(x)|^2 \, dx$ as small as possible.

Axler Note. The polynomial 6.65 is an excellent approximation to the answer to this exercise, but here you are asked to find the exact solution, which involves powers of π . A computer that can perform symbolic integration should help.

Solution. Similar to Exercise 6C17, we project $\sin x$ onto $\text{span}(1, x, x^2, x^3, x^4, x^5)$ in $C[-\pi, \pi]$. Note that $\sin x$ is orthogonal to $1, x^2$, and x^4 on $[-\pi, \pi]$ because it is an odd function. Hence, we only need to consider the projection onto $\text{span}(x, x^3, x^5)$. Using the Gram–Schmidt procedure, we can find an orthonormal basis of $\text{span}(x, x^3, x^5)$ and then compute the projection of $\sin x$ onto this subspace. Calculating the orthogonal basis, we begin with $f_1 = x$ with $\|f_1\|^2 = \frac{2\pi^3}{3}$. Then

$$f_2 = x^3 - \frac{\langle x^3, f_1 \rangle}{\|f_1\|^2} f_1 = x^3 - \frac{3\pi^2}{5} x, \quad \|f_2\|^2 = \frac{8\pi^7}{175}.$$

Finally,

$$f_3 = x^5 - \frac{\langle x^5, f_1 \rangle}{\|f_1\|^2} f_1 - \frac{\langle x^5, f_2 \rangle}{\|f_2\|^2} f_2 = x^5 - \frac{10\pi^2}{9} x^3 + \frac{5\pi^4}{21} x, \quad \|f_3\|^2 = \frac{128\pi^{11}}{43659}.$$

Projecting $\sin x$ onto $U = \text{span}(x, x^3, x^5)$ gives

$$P_U \sin x = \frac{\langle \sin x, f_1 \rangle}{\|f_1\|^2} f_1 + \frac{\langle \sin x, f_2 \rangle}{\|f_2\|^2} f_2 + \frac{\langle \sin x, f_3 \rangle}{\|f_3\|^2} f_3,$$

where the inner products are computed as follows:

$$\begin{aligned} \langle \sin x, f_1 \rangle &= \int_{-\pi}^{\pi} x \sin x \, dx = 2\pi, \\ \langle \sin x, f_2 \rangle &= \int_{-\pi}^{\pi} \left(x^3 - \frac{3\pi^2}{5} x \right) \sin x \, dx = \frac{4\pi}{5} (\pi^2 - 15), \\ \langle \sin x, f_3 \rangle &= \int_{-\pi}^{\pi} \left(x^5 - \frac{10\pi^2}{9} x^3 + \frac{5\pi^4}{21} x \right) \sin x \, dx = \frac{2}{63} (8\pi^5 - 840\pi^3 + 7560\pi). \end{aligned}$$

The coefficients of f_1, f_2 , and f_3 in the projection are then

$$\frac{\langle \sin x, f_1 \rangle}{\|f_1\|^2} = \frac{3}{\pi^2}, \quad \frac{\langle \sin x, f_2 \rangle}{\|f_2\|^2} = \frac{35(\pi^2 - 15)}{2\pi^6}, \quad \frac{\langle \sin x, f_3 \rangle}{\|f_3\|^2} = \frac{693(\pi^4 - 105\pi^2 + 945)}{8\pi^{10}}.$$

Hence, the polynomial $p(x) = P_U \sin x$ that minimizes the integral is $p(x) = \alpha x + \beta x^3 + \gamma x^5$, where

$$\begin{aligned}\alpha &= \frac{105}{8\pi^6}(\pi^4 - 153\pi^2 + 1485), \\ \beta &= -\frac{315}{4\pi^8}(\pi^4 - 125\pi^2 + 1155), \\ \gamma &= \frac{693}{8\pi^{10}}(\pi^4 - 105\pi^2 + 945).\end{aligned}$$

■

Exercise 6C19

Suppose V is finite-dimensional and $P \in \mathcal{L}(V)$ is an orthogonal projection of V onto some subspace of V . Prove that $P^\dagger = P$.

Solution. For $u, v \in V$, write $u = u_U + u_{U^\perp}$ and $v = v_U + v_{U^\perp}$ with $u_U, v_U \in U = \text{range } P$ and $u_{U^\perp}, v_{U^\perp} \in U^\perp = \text{null } P$. Then $Pu = u_U$ and $Pv = v_U$, so

$$\langle Pu, v \rangle = \langle u_U, v_U + v_{U^\perp} \rangle = \langle u_U, v_U \rangle$$

since $u_U \in U$ and $v_{U^\perp} \in U^\perp$. On the other hand,

$$\langle u, Pv \rangle = \langle u_U + u_{U^\perp}, v_U \rangle = \langle u_U, v_U \rangle$$

since $u_{U^\perp} \in U^\perp$ and $v_U \in U$. Hence, $\langle Pu, v \rangle = \langle u, Pv \rangle$ for all $u, v \in V$, so $P^\dagger = P$. ■

Exercise 6C20

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Show that

$$\text{null } T^\dagger = (\text{range } T)^\perp \quad \text{and} \quad \text{range } T^\dagger = (\text{null } T)^\perp.$$

Solution. Let P be the orthogonal projection of W onto $\text{range } T$ and let $\tilde{T} : (\text{null } T)^\perp \rightarrow \text{range } T$ denote the restriction of T to $(\text{null } T)^\perp$, so that $T^\dagger = \tilde{T}^{-1}P$. Since $\tilde{T}^{-1} : \text{range } T \rightarrow (\text{null } T)^\perp$ is a bijection, and P maps W onto $\text{range } T$,

$$\text{null } T^\dagger = \text{null}(\tilde{T}^{-1}P) = \text{null } P = (\text{range } T)^\perp,$$

where the middle equality uses injectivity of $\tilde{T}^{-1}$, and

$$\text{range } T^\dagger = \tilde{T}^{-1}(P(W)) = \tilde{T}^{-1}(\text{range } T) = (\text{null } T)^\perp,$$

where the second equality uses surjectivity of P onto $\text{range } T$. ■

Exercise 6C21

Suppose $T \in \mathcal{L}(\mathbb{F}^3, \mathbb{F}^2)$ is defined by

$$T(a, b, c) = (a + b + c, 2b + 3c).$$

- (a) For $(x, y) \in \mathbb{F}^2$, find a formula for $T^\dagger(x, y)$.
- (b) Verify that the equation $TT^\dagger = P_{\text{range } T}$ from 6.69(b) holds with the formula for $T^\dagger$ obtained in (a).
- (c) Verify that the equation $T^\dagger T = P_{(\text{null } T)^\perp}$ from 6.69(c) holds with the formula for $T^\dagger$ obtained in (a).

Solution.

- (a) From $a + b + c = 2b + 3c = 0$, we see that $\text{null } T = \text{span}(1, -3, 2)$, so

$$(\text{null } T)^\perp = \{(a, b, c) \mid a - 3b + 2c = 0\}.$$

Moreover, $\text{range } T = \mathbb{F}^2$, so $P_{\text{range } T} = I$. Let $T^\dagger(x, y) = (a, b, c)$. Then $T(a, b, c) = (x, y)$ and $(a, b, c) \in (\text{null } T)^\perp$. Hence, we have the system of equations

$$\begin{cases} x = a + b + c, \\ y = 2b + 3c, \\ 0 = a - 3b + 2c. \end{cases}$$

This yields the solution

$$T^\dagger(x, y) = \left(\frac{13x - 5y}{14}, \frac{3x + y}{14}, \frac{-2x + 4y}{14} \right).$$

- (b) For $(x, y) \in \mathbb{F}^2$, we have

$$TT^\dagger(x, y) = T\left(\frac{13x - 5y}{14}, \frac{3x + y}{14}, \frac{-2x + 4y}{14}\right) = (x, y),$$

as required.

- (c) We have $P_{(\text{null } T)^\perp}$ being the projection onto the plane $a - 3b + 2c = 0$. Hence,

$$P_{(\text{null } T)^\perp}(a, b, c) = (a, b, c) - \frac{a - 3b + 2c}{14}(1, -3, 2) = \left(\frac{13a + 3b - 2c}{14}, \frac{3a + 5b + 6c}{14}, \frac{-2a + 6b + 10c}{14} \right).$$

For $(a, b, c) \in \mathbb{F}^3$, we have

$$T^\dagger T(a, b, c) = T^\dagger(a + b + c, 2b + 3c) = \left(\frac{13a + 3b - 2c}{14}, \frac{3a + 5b + 6c}{14}, \frac{-2a + 6b + 10c}{14} \right),$$

as required. ■

Exercise 6C22

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$TT^\dagger T = T \quad \text{and} \quad T^\dagger TT^\dagger = T^\dagger.$$

Axler Note. Both formulas above clearly hold if T is invertible because in that case we can replace $T^\dagger$ with T^{-1} .

Solution. Let $V = \text{null } T \oplus (\text{null } T)^\perp$ so that any $v \in V$ can be written as $v = u + w$ with $u \in \text{null } T$ and $w \in (\text{null } T)^\perp$. For the first equality, we have $Tv = Tw \in \text{range } T$. Then $T^\dagger Tw = w$ since $T^\dagger$ restricted to $\text{range } T$ is the inverse of T restricted to $(\text{null } T)^\perp$. Hence,

$$TT^\dagger Tv = TT^\dagger Tw = Tw = Tv$$

for all $v \in V$, so $TT^\dagger T = T$.

For the second equality, let $w \in W$ and write $w = x + y$ with $x \in \text{range } T$ and $y \in (\text{range } T)^\perp$. Then $T^\dagger w = T^\dagger x \in (\text{null } T)^\perp$, so

$$T^\dagger TT^\dagger w = T^\dagger T(T^\dagger x) = T^\dagger x = T^\dagger w.$$

Hence, $T^\dagger TT^\dagger = T^\dagger$. ■

Exercise 6C23

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$(T^\dagger)^\dagger = T.$$

Axler Note. The equation above is analogous to the equation $(T^{-1})^{-1} = T$ that holds if T is invertible.

Solution. By Exercise 6C20, $\text{null } T^\dagger = (\text{range } T)^\perp$ and $\text{range } T^\dagger = (\text{null } T)^\perp$. Applying 6.68 to $T^\dagger$ in place of T and substituting via Exercise 6C20, for $v \in V$,

$$(T^\dagger)^\dagger v = (T^\dagger|_{(\text{null } T^\dagger)^\perp})^{-1} P_{\text{range } T^\dagger} v = (T^\dagger|_{\text{range } T})^{-1} P_{(\text{null } T)^\perp} v.$$

By 6.67, $T|_{(\text{null } T)^\perp}$ is an isomorphism onto $\text{range } T$ whose inverse is $T^\dagger|_{\text{range } T}$, so $(T^\dagger|_{\text{range } T})^{-1} = T|_{(\text{null } T)^\perp}$. Hence, $v - P_{(\text{null } T)^\perp} v \in \text{null } T$,

$$(T^\dagger)^\dagger v = T(P_{(\text{null } T)^\perp} v) = Tv. \quad \text{■}$$